

ICST INETRDISCIPLINARY CENTER FOR SMART TECHNOLOGIES

Center of CS Department's Software Engineering Track

SmartRail

Know where to board.

SmartRail monitors train-seat availability and shows passengers which cars have more free seats.

SENSE IR sensors + ESP32 detect occupied seats.

GUIDE A browser display highlights available cars.

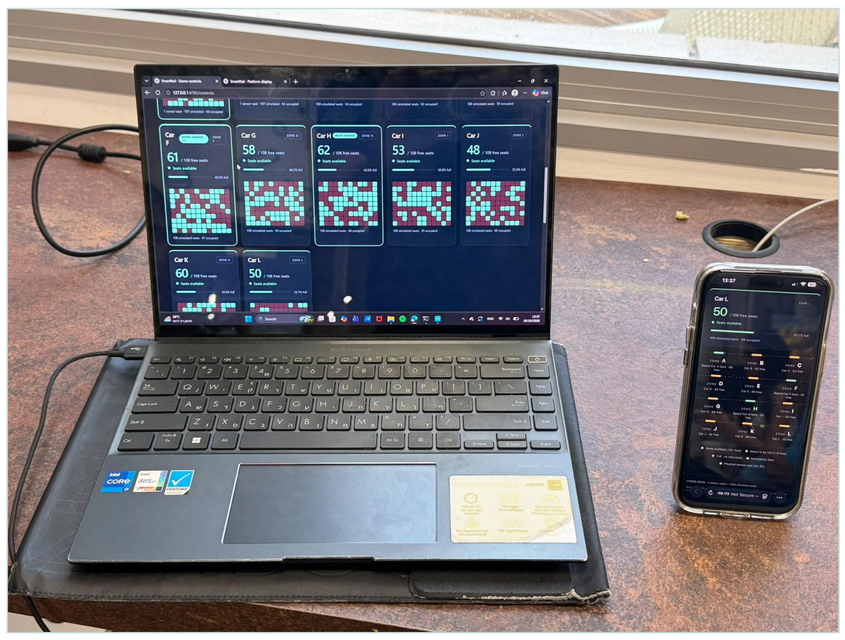
SCALE 2 physical inputs + 1,294 simulated seats.

12 virtual cars | 108 seats per car | 1,296 total

THE PROTOTYPE

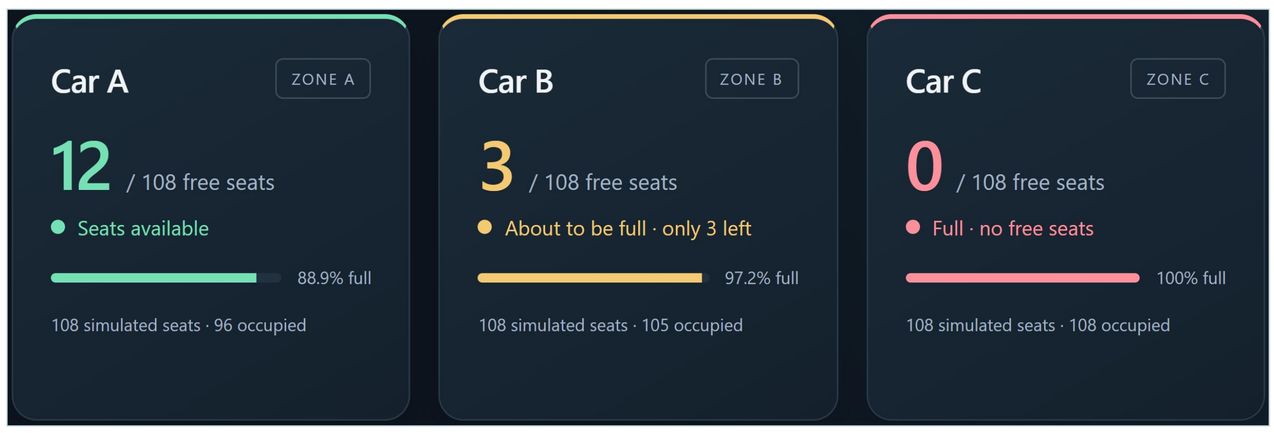


Two seat models, ESP32 and live interfaces.



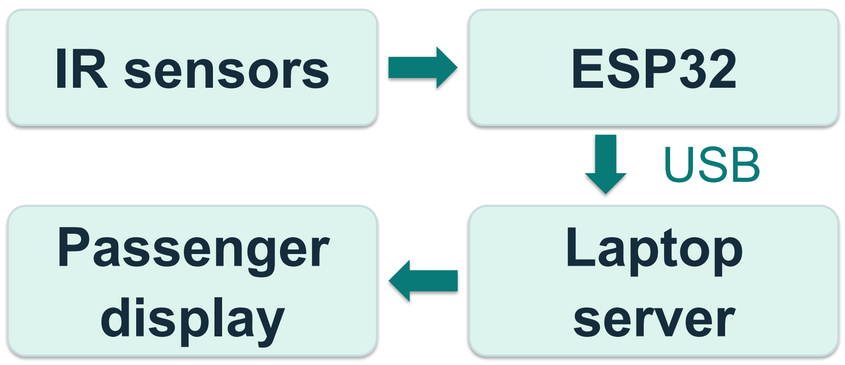
Laptop controls; phone display.
Viewing over local Wi-Fi;
sensor link remains USB.

PASSENGER VIEW



Simulation example: free seats by car.
Unknown readings are never counted free.

HOW IT CONNECTS



Simulated seats also feed the server.
Availability guidance, not reservations.

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github.com/abedazem13/SmartRail

A Project in Internet of Things (IoT)

Computer Science Department

הטכניון מכון טכנולוגי לישראל